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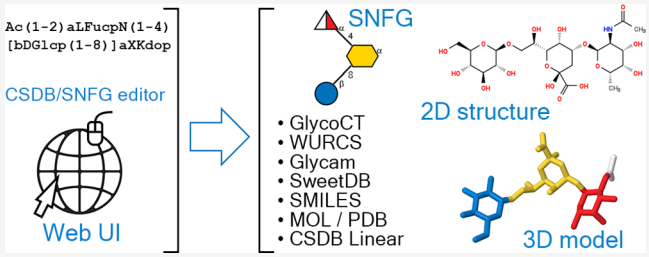
CSDB/SNFG Structure Editor: An Online Glycan Builder with 2D and 3D Structure Visualization

Andrei Y. Bochkov and Philip V. Toukach*

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ABSTRACT: This article describes features, usage, and application of an CSDB/SNFG Structure Editor, a new online tool for quick and intuitive input of carbohydrate and derivative structures using Symbol Nomenclature for Glycans (SNFG). The Editor is built on a platform of the Carbohydrate Structure Database (CSDB) and relies on its online services via the dedicated web-API. The Editor allows building of oligo- and polymeric glycan structures and supports most features of natural glycans, such as underdetermined structures, alternative branches, repeating subunits, SMILES specification of atypical monomers, and others. The vocabulary of building blocks contains 600+ monomeric residues, including 327 monosaccharides. Support for SMILES allows input and visualization of chemical structures of virtually unlimited complexity. On the other hand, the interface follows the recognized GlycanBuilder style easy to novice users. The export feature includes support for CSDB Linear, GlycoCT, WURCS, SweetDB, and Glycam notations, SMILES codes, MOL/PDB atomic coordinate formats, raster and vector SNFG images, and on-the-fly visualization as 2D structural formulas and 3D molecular models. Integration of the Editor into any web-based glycoinformatics project is straightforward and simple, similarly to any other modern JavaScript application.



INTRODUCTION

Since the introduction of computational methods in glycochemistry, a number of software tools to prepare graphical representations of glycan structures and their export to various carbohydrate notations has been developed. These tools and their capabilities have been recently reviewed in detail.[1]

From our viewpoint, several criteria making the glycan builder tool convenient for usage and versatile applications can be named:

- the visual editing principle when the graphical representation of data is automatically synchronized with user action, such as appending, removal, or update of items;
- exportable graphical sketching using SNFG notation, a modern standard for glycan depiction;[2,3]
- export of a structure into computer-readable text notations,[4] as a basic function making the tool useful beyond graphical depiction and allowing integration with carbohydrate databases and computational tools;
- seamless import and export to/from at least one data format, so that the created structure could be saved, shared over a network, or restored in the other instance of the editor.

To date, we have found five glycan builders that meet the above-mentioned criteria: SugarSketcher,[5] GlycoGlyph,[6] GlycanBuilder2,[7] Draw RingS,[8] and Polys Glycan Builder.[9] Each of these tools has limitations in user input and depiction of specific structural features. None of them supports input of alternative branches at the specified inline or terminal location (however, in GlycanBuilder2 it can be done via fragments with undefined attachment) and export of structure into atom-level formats. Input of repeated subunits in the user interface (in contrast to text import) is supported in SugarSketcher and GlycanBuilder2, while nesting of repeated units is supported by GlycanBuilder2 only. Input of basic structural uncertainties (unknown anomeric/absolute configurations, cyclization type, and linkage position) is completely implemented in SugarSketcher, GlycanBuilder2, and GlycoGlyph. All five support residue superclasses, and among more advanced uncertainty features only GlycanBuilder2 supports probability annotations for linkages and glycan fragments. Chemical validation of user input has not been fully implemented in these editors, thus leaving a possibility to assemble chemically invalid structures (e.g., linkage via a nonexistent 6-O atom in fucose, attachment of more than one substituent to the same position in an acceptor, etc.), which yield invalid nonprocessable output on export. From the usability point of view, only GlycanBuilder2 follows a typical interface layout of a chemical structure editor with menu,

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Table 1. Comparison of the Functionality of the Selected Existing Glycan Builders and CSDB/SNFG Editor

function/feature[a]	CSDB/SNFG Editor	Glycan Builder2	Sugar Sketcher	Glyco Glyph	Draw RingS	Polys Glycan Builder
visual editing	+	+	+	+	+	+
colored SNFG (interface and export to image file)	+	+	+	+	+	+
export/import of text notation(s)	+	+	+	+	+	+
monosaccharide vocabulary (no. of items)	300+	70+	70+	80+	70+	100+
nonmonosaccharide residue vocabulary (no. of items)	300+	30+	20+	<10	−	−
input of nonvocabulary residues	+[b]	−	−	+[c]	−	−
residue superclasses	+	+	+	+	+	+
simple uncertainties (absolute and anomeric configurations, cyclization type, linkage positions)	+	+	+	+	+/−	+/−
probabilistic linkages	−	+	−	−	−	−
alternative branches	+	+/−[d]	−	−	−	−
repeating units	+	+	+	−	−	−
nested repeating units	+	+	−	−	−	−
chemical validation of input structure	+	−	−	−	−	−
2D visualization as a structural formula	+	−	−	−	−	−
3D visualization as a molecular model	+[e]	−	−	−	−	+

[a]Levels of feature support: +, fully supported; +/−, partially supported; −, not supported. [b]Specified by chemical name and/or SMILES. [c]Specified by chemical name. [d]Input of fragments with undefined attachment. [e]Including SNFG coloring of residues.

toolbars, collection of structural templates, and the drawing sketch (top to bottom). SugarSketcher utilizes a novel interface scheme, however, not accepted by all users. Other tools are designed in a web-page style (HTML form elements and a sketch). Graphical representation strictly complies with shapes and colors of SNFG specification in GlycanBulder2 and GlycoGlyph. In summary, GlycanBuilder2 seems to be a most feature-rich and easy to use glycan editing tool, but it still lacks valuable functionality. Its main drawback is implementation as a Java application. Since use of Java is strongly discouraged in the modern web due to security concerns and support for Java applets has been discontinued by a majority of browsers, integration of GlycanBuilder2 into web-based projects has become hampered.

Aiming at the enhancement of functionality of glycan builders, we developed CSDB/SNFG structure editor, a cross-platform online application for glycan input and editing. The functionality and features of the above five glycan builders and new CSDB/SNFG structure editor are compared in Table 1. The Editor is built on a platform of the Carbohydrate Structure Database (CSDB[10]) and relies on its online services via the dedicated web-API. We followed the well-recognized interface style that has been accepted by users since early versions of GlycanBuilder[11] and extended it according to user demands and support of the novel structural features. We supplemented the sketched structures in carbohydrate notation by atom-level information, thus linking a glycan builder to the molecular modeling module of CSDB. It allowed users to work with optimized and residue-assigned three-dimensional structures of glycans directly in a browser.

FEATURES

The key features of the CSDB/SNFG Editor include the following:

- following the SNFG specification for visualization;
- fluent glycan sketching (one click per residue in the simplest cases);
- a simplified *novice* sketching mode for newcomers and an *expert* mode for experienced users;
- instant visualization as a chemical structural formula;
- generation of the optimized 3D molecular model;
- native support of CSDB Linear notation (graphical input and export/import of all notation features);
- usage of an extensive regularly updated monomeric vocabulary of CSDB (600+ entries);
- chemical validation of the sketched structure on the fly (in most cases, a chemically impossible structure cannot be entered; in borderline cases, the user is warned);
- export to GlycoCT condensed format with alternative substructures, repeating subunits, and noncarbohydrate moieties supported;
- export to other carbohydrate notations (WURCS, Glycam), chemical structure (SMILES), and atomic coordinate formats (MOL and PDB with native or GLYCAM residue naming).

Structure Model and Native Notation. The native carbohydrate notation internally used by the Editor is the CSDB Linear.[12,13] A glycan structure (a tree graph) in the Editor is built up of elementary entities (nodes, *residues* in terms of CSDB Linear). Residues are connected by bonds (graph edges). The vocabulary of entities is loaded upon every run of the Editor in a client browser and includes all residues available in the CSDB at the moment. By June 2021, the CSDB monomeric namespace included 327 monosaccharide and alditol units, 19 residue superclasses, 119 simple monovalent substituents (carboxylic acids, alcohols, etc.), and 215 polyvalent non-monosaccharide units (inositols, amino acids, hydroxylated fatty acids, etc.). In the Editor workflow and further in this text, the residue vocabulary is divided into two groups of entities: *residues* (polyvalent units, monosaccharides and others) and *modifications* (monovalent nonmonosaccharide units). Since modifications are not substitutable (being monovalent by definition), they can be depicted as lists attributed to their carrier residues rather than separate nodes, which saves space on the drawing canvas. CSDB residues have a defined chemical structure with possible variations depending on up to three descriptors: absolute configuration, ring size, and anomeric configuration. All three are attributed to monosaccharides where applicable, and only the absolute configuration is attributed to alditols and chiral noncarbohydrate constituents. A residue encoding scheme and

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possible descriptor values in CSDB Linear notation are summarized in Figure 1.

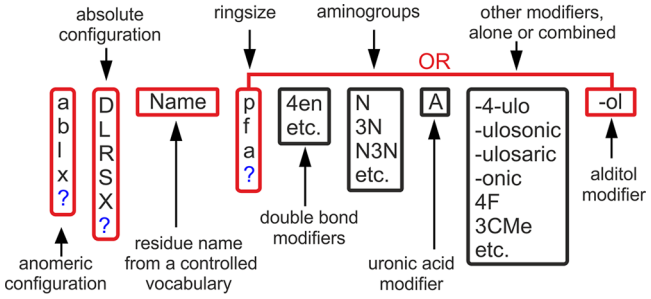


Figure 1. Encoding of a residue in CSDB Linear. Obligatory fields are outlined in red. Reprinted with permission from ref 13. Copyright 2020 American Chemical Society.

The Editor allows input and depiction of the following features of the CSDB Linear notation:

- all noncyclic oligomeric and polymeric topologies;
- topologies that combine repeating and nonrepeating parts (including nested subunits of both types);
- CSDB namespace of carbohydrate and noncarbohydrate monomers;
- uncertainties at the level of residue identity, substitution pattern, and residue configurations;
- uncertainties at the topological level: sets of alternative branches, variable count of repeats;
- aliases for arbitrary (nonvocabular) residues with SMILES specification of the structure;
- regular (glycosidic, amidic, diester, and other bonds with elimination of water), chelate (with elimination of two water molecules), and carbon−carbon linkages.

Interface. The user interface of the Editor is divided into two major sections—the *Drawing Section* and the *Previews Section*. The interface of the Drawing Section (Figure 2, top) was designed to resemble a classical chemical structure editor with a built-in library of templates and includes a control button bar (1), a residue toolbar with sub-bars grouped as tabs (2), a group of switches for major settings (3), and lookup fields for search of less abundant residues/modifications (4). Drawing canvas (5) is located below the toolbars and displays the SNFG image of a glycan structure. A status bar at the bottom of Drawing Section (6) displays a CSDB Linear code of the sketched structure.

Sketching Process. The glycan structure is created by adding residues to the drawing canvas one by one. The sketching process was designed to be as prompt as possible. Popular constituents are instantly accessible as SNFG-marked buttons on the toolbar and can be appended to the structure in a single-click manner. To facilitate a glycan building process, each residue appears on the sketch with preassigned default descriptors: absolute configuration and ring size (for monosaccharide residues) are set to the most naturally abundant for this residue; anomeric configuration of monosaccharides is set to unknown (*?*). If the desired value differs from the default one, it can be easily adjusted by clicking the dedicated control displayed as a semitransparent disk over the SNFG icon (Figure 3, arrows 1−3). On click, these controls sequentially loop through all possible descriptor values. These values are specific for a particular residue, depending on its chemical structure and current status of other descriptors, thus ruling out the chemically nonsense input (e.g., setting a pyranose ring for a tetrose or assigning an anomeric configuration to monosaccharide with ring size set to open form). The newly added residue is automatically connected with the selected (embordered in red) residue by the linkage line, if present. A newly formed linkage gets descriptors (outgoing and ingoing positions) assigned to defaults in a similar manner: default outgoing position depending on the type of donor residue and unknown further-editable ingoing position in the acceptor residue (Figure 3, arrow 4). The most recently added residue becomes a selected one automatically, thus allowing immediate chain elongation (Figure 3, arrow 5). The majority of visual glycan building tools require preconfiguration of the residue and linkage properties before a new item is appended to the sketched structure. Since configurations are typically designed as a group of dropdowns and/or radio-buttons, it requires more clicks to go through for every residue added. We hope a single-click sketching workflow in the CSDB/SNFG Editor simplifies the glycan building process as compared to a preconfiguration approach.

Novice and Expert Modes. The Editor has two operational modes selectable by a control switch on the toolbar. In the *Expert mode* all features are enabled. The *Novice mode* simplifies the interface and has the following limitations:

- only the 127 most abundant residues and modifications can be drawn, and complete lists of residues and modifications are hidden;
- absolute configurations, ring sizes, and outgoing bond positions are assumed to be the default ones for each residue and cannot be edited;
- rarely occurring bonds (amino/amide → amino/amide, amino/amide → hydroxy, hydroxy → amino/amide) cannot be drawn.

Previews and Visualization. The Previews Section (Figure 2, bottom) is located below the Drawing Section. Any change of the glycan structure on the sketch triggers generation of the corresponding CSDB Linear code. The generated code is immediately validated by the server-side parsing service using a dedicated web-API. If any issues are detected by the parser, the error message(s) appears at the top of the Previews Section. A valid glycan structure is visualized on-the-fly by the server-side generator of the SNFG image (GIF) (7), which can be further exported in high resolution on a solid or transparent background. To balance the load on the server web-API, the other previews are updated after a timeout (5 s by default) elapses since the last change was applied to the structure. They include structure in GlycoCT (8) and SweetDB (9) format and a planar (2D) structural formula with stereochemistry (10). For glycans containing uncertainties in a primary structure, a list of possible fully determined structures (in CSDB Linear notation) is displayed for selection; the first item is preselected and immediately visualized as a chemical structure. On user request (*Show 3D* button), the displayed 2D structure is converted to a 3D model (11). If multiple 3D structures are possible for a single 2D structure (e.g., with different anomericity), another selection list is displayed. A 3D model is generated by the REStLESS service[14] on the basis of predominant cycle conformations and MMFF single-point optimization. Equipping of REStLESS with a module to utilize the inter-residue bridge conformation data from molecular dynamics calculations is a question for the near future. A 3D model in MOL format is transferred for display to a JSMol JavaScript module,[15] providing wide features of molecule visualization. Atoms within the model are colored correspondingly to SNFG color of the residue which they belong to. The

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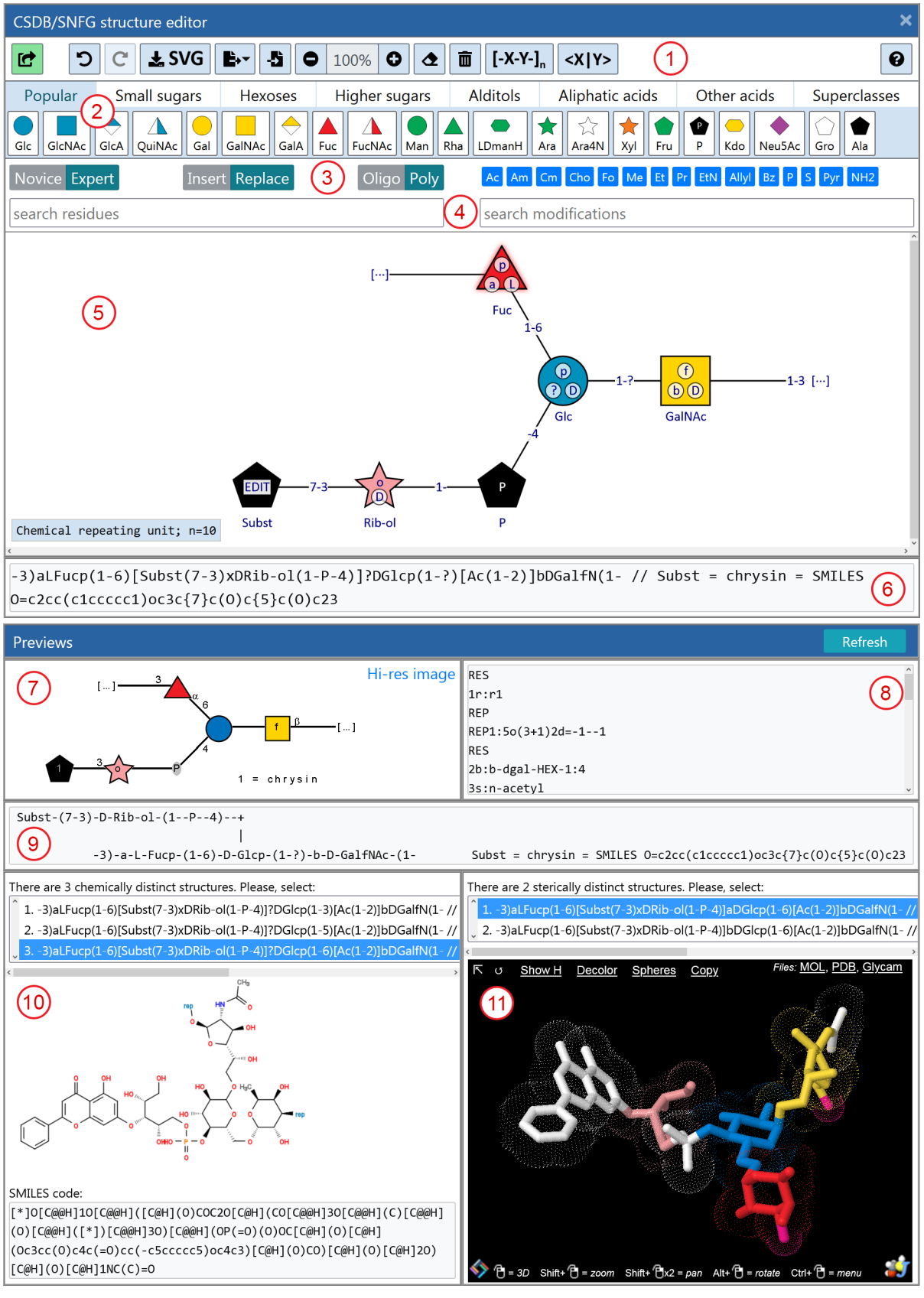


Figure 2. CSDB/SNFG Structure Editor interface.

JSMol window is appended with a few rendering mode and data export shortcuts; the rest of its functionality is available from a right-click menu.

Import/Export. The drawn glycan can be exported to notations for which translators are available at the CSDB platform. Currently, they include general chemical formats

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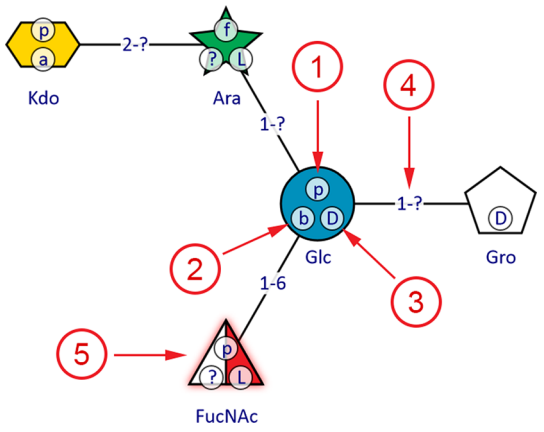


Figure 3. Example of sketched glycan. Residue (1−3) and linkage (4) controls and current residue highlight (5) are pointed with red arrows and numbers (see text). Corresponding CSDB Linear code: `Ac(1−2)?LFucpN(1−6)[aXKdop(2-?)?LAraf(1-?)]bDGlcp(1-?)xDGro`.

(SMILES, MOL, PDB with two types of ligand naming) and a number of text carbohydrate notations. Not all of the notations support all features of glycan structure that can be encoded in CSDB Linear, thus limiting output in some cases. The Editor is capable of importing structures from the CSDB Linear (via URL or via import form) and GlycoCT condensed (via import form) notations. Supported export and import formats are listed in Table 2. Examples of various sketched structures and corresponding exported CSDB Linear and GlycoCT strings are provided in the Supporting Information, Table S1.

Compatibility with GlycoCT. The Editor features comprehensive output compatibility with the GlycoCT condensed format thanks to a new CSDB Linear to GlycoCT converter developed within the CSDB project simultaneously with the Editor. Any valid structure drawn in the Editor and hence supported by its native CSDB Linear notation is convertible to the GlycoCT condensed string. To date, this is the only translator implementing full support of alternatives (ALT), subrepeats (REP, including nested ones), and non-monosaccharide (NON) units in the GlycoCT notation, as documented in the original publication on GlycoCT.[16]

Table 2. Import and Export Features of the CSDB/SNFG Editor

notation or object	data format	export	import
CSDB Linear	string	+	+
GlycoCT condensed[16]	multiline string	+	+[a]
SweetDB[17]	multiline string	+	
WURCS 2.0[18]	string	+[b]	
Glycam[19]	string	+[b,c,d]	
SMILES[20]	string	+	
2D structural formula	PNG	+[e]	
3D structure	PDB with classic residue naming	+	
	PDB with Glycam residue naming	+	
	MOL	+	
	LiteMOL object model[21]	+	
	VRML, STL, X3D	+[f]	
3D structure snapshot	GIF, JPG, PNG	+[f]	
SNFG	GIF	+	
	SVG	+	

[a]Alternative, repeated, and nonmonosaccharide units are not supported on import (except polymers, when the whole structure is a repeated unit). [b]Subrepeats are converted to regular oligomers; large repeating subunit count is reduced to $n = 5$. [c]*Subst* aliases are not supported. [d]Alternative branches are not supported. [e]*Save image as* command in a browser context menu. [f]*File* > *Export* command in JSMol applet context menu.

It should be noted that encoding of nonmonosaccharide moieties is described in a limited way and has very few examples in otherwise detailed GlycoCT online specification.[22] Generally, GlycoCT notation assumes that nonmonosaccharide constituents are encoded as *substituents* (this is the case for a defined set of small molecules most often directly bound to monosaccharides) or as NON-items (all other ones not listed as substituents). The original GlycoCT publication[16] listed 37 small molecule substituents; subsequently, this list was supplemented by various researchers who used GlycoCT. For CSDB Linear to GlycoCT converter we took a list maintained by the Glytoucan project[23] as a reference (see https://glycosmos.org/glytoucans). In the translation operations those non-monosaccharide entities from CSDB vocabulary, for which a corresponding entry exists in this reference list, are encoded as substituents and otherwise as NON-items.

The GlycoCT specification suggests the usage of InChI[24] to characterize a nonmonosaccharide entity under the *SmallMolecule* keyword within a NON item. Since CSDB Linear uses SMILES[20] to store a chemical structure of monomeric units, we introduced an extension for GlycoCT NON-item to support SMILES, thus making the notation more versatile. We suggested usage of a SMILES code prefixed with a "SMILES" keyword and a blank space as a second option in the *SmallMolecule* field. This approach allows one to distinguish InChI and SMILES input easily both for a human reader and for any computer parser. In addition, we introduced an optional field *Description* for NON-item to store the chemical name of the unit and/or further human-readable annotations besides the structure itself (Figure 4A and 4B).

CSDB Linear does not limit the structure topology in regard to residue class, e.g., a chain of nonmonosaccharide residues can be a terminal fragment on a glycan or it can be further substituted with monosaccharides. GlycoCT format does not declare the possibility for substitution on the NON-item either with a monosaccharide or with another NON-item. Documented fields of the NON-item imply input of the outgoing linkage only (*Parent* field for numeric id of an acceptor residue and *Linkage* field for string representation of outgoing linkage) but not the ingoing ones. To address this gap and provide full compatibility of CSDB Linear (and hence CSDB/SNFG Editor) with GlycoCT on the topological level, we implemented an extension to GlycoCT, i.e., referencing of nonmonosaccharide units (NONs) and their linkages in RES and LIN sections, respectively. All NONs are indexed together with units of other types within the RES section using a type identifier "n" (noncarbohydrate unit). "n"-type has been already declared in the GlycoCT specification along with "conventional" types (b, base type; s, substituent; r, repeating unit; a, alternative unit). Indexing of NONs in the RES section allows one to encode all of their linkages (in- and outgoing) in the LIN section with reference to their indices. SMILES codes of residues to be stored in the NON-item are provided by CSDB services and feature atom numbering. Referencing SMILES atom numbers upon encoding positions of linkages to/from NON-items allows an unambiguous description of a chemical structure. An example in Figure 4C shows encoding of a chain of nonmonosaccharides

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A.
```
NON1
Parent:2
Linkage:o(-1+1)n
SmallMolecule:InChI=1S/C4H8O3/c1-3(5)2-4(6)7/h3,5H,2H2,1H3,(H,6,7)
```

B.
```
NON1
Parent:2
Linkage:o(-1+1)n
SmallMolecule:SMILES [4CH3][3CH](O)[2CH2][1C](=O)O
Description:3-hydroxybutanoic acid
```

C.
```
RES
1b:x-dgal-HEX-1:5
2n:n1
3n:n2
4b:b-lgal-HEX-1:5|6:d
5s:n-acetyl
LIN
1:1o(4+1)2n
2:2n(3+1)3n
3:3n(2+1)4d
4:4d(2+1)5n
NON
NON1
Parent:1
Linkage:o(4+1)n
SmallMolecule:SMILES [4CH3][3C@H](O)[2CH2][1C](=O)O
Description:(S)-3-hydroxybutanoic acid
NON2
Parent:2
Linkage:n(3+1)n
SmallMolecule:SMILES [3CH3][2C@@H](O)[1C](=O)O
Description:(R)-lactate
```

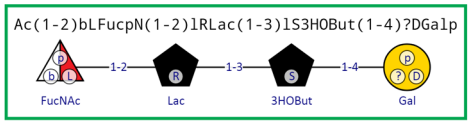


Figure 4. Implemented extensions to current GlycoCT condensed specification.[22] (A and B) Example of 3-hydroxybutanoic acid with unknown stereochemistry represented as a NON item using current (A) and suggested (B) GlycoCT specification. (C) Example of a structure consisting of two monosaccharides linked via two non-monosaccharide residues. Supplemented GlycoCT condensed notation is provided. Green box displays the structure in CSDB/SNFG Editor drawing canvas and in CSDB Linear notation.

capped with monosaccharide units in the supplemented version of GlycoCT. A special case occurs when a polyvalent CSDB residue has to be translated into a GlycoCT s-type unit (substituent). In this case, further substitution is chemically possible (e.g., on a hydroxy function of a carboxyl-linked lactic acid residue) and hence allowed in the CSDB Linear notation (and CSDB/SNFG Editor). Since GlycoCT forbids substituents to be substituted, nonterminal noncarbohydrate units are always encoded as n-type (see encoding of lactate and its linkages in Figure 4C). Examples of GlycoCT output for various structures are provided in the Supporting Information, Table S1.

USAGE

Residue Input. Two methods for residue input are available: via a click on a toolbar from a corresponding tab (contains the most abundant residues) and via a lookup field *Search residues*. Lookup fields operate as filters displaying a dropdown list of entities where a chemical name or abbreviation partially matches the search string. Clicking the row in the lookup dropdown adds a corresponding residue to the sketch. If the sketch contains residues, one of them is always selected (highlighted with a red halo). Depending on the state of the *Insert/Replace* switch, a new residue is appended as a donor to the selected one or replaces it. The focus is automatically shifted to the recently added residue. Any residue can be selected manually by clicking its SNFG icon on the sketch.

Modification Input. Input of modifications is implemented similarly to residues—from the toolbar or using a lookup field *Search modifications*. Unlike residue input, an additional step to specify a modification attachment position is required. When a modification is clicked on the toolbar or selected from a lookup dropdown list a Modification dialogue (Figure 5A) is displayed. A modification of interest is preselected and can be changed by clicking on another one or via a search in the complete list. For selection of the attachment position in the acceptor residue, only positions vacant at the moment are listed. The C−C bond checkbox switches the attachment position list to display unoccupied carbons. Two input options are available: clicking the *Append as modification* button adds the modification to the target residue; clicking the *Append as residue* button adds a selected entity as a new separate residue on the sketch (a donor of the current residue). Added modifications are displayed in the bottom of the dialogue and can be removed with a delete button. The dialogue can be alternatively activated by clicking the residue caption consisting of its abbreviation and modifications (if any) prefixed with their positions.

Bond Property Input. Bond properties are edited in the Bond dialogue (Figure 5B). To open the dialogue click on a bond caption. The bond under edition is highlighted by bold linkage and a violet glow. Depending on the linked residues, corresponding lists of available linkage positions are displayed. Only vacant positions (not occupied by other bonds and modifications) are listed. Acceptor linkage is a bonded position in the acceptor residue (usually on the right). Donor linkage is a bonded position in the donor residue (usually on the left); for monosaccharide residues an anomeric atom is preselected.

Additional controls include the following:

- a carbon−carbon bond checkbox (for input of C-glycosides or atypical monosaccharides with branched carbon skeleton);
- a dual-bond checkbox is displayed if the selected donor position can form a chelate linkage with two positions in the acceptor (e.g., carbonyl atom prone to diacetal formation). With this checkbox checked, two acceptor positions have to be selected sequentially.

Atypical Substituent Input. Atypical residues not included in the CSDB monomer namespace use a generic alias *Subst*; a corresponding button is located on the Superclasses tab. To open a dialogue to edit specific properties of a *Subst* residue (Figure 5C), a clickable *EDIT* control is displayed in front of the *Subst* residue icon on the sketch. The editable *Subst* properties include the user description (usually, chemical name), SMILES code, and substitutable positions (comma-separated list). Substitutable positions are used in the Bond dialogue to specify linkages connecting *Subst* with other residues.

To make assembly of the chemical structure from the residue-based code possible, atoms connected with other residues must be supplemented with numbering in the SMILES code (number in curly brackets preceding the atom symbol in SMILES, according to CSDB Linear notation rules).[13]

Polymeric Structure Input. The sketched structure can be interpreted as an oligomer or as a regular polymer repeating unit. The latter case implies one more bond between the repeating units, which is displayed as two open bonds at the left and right termini of the structure. The [···] mark indicates the next/previous repeating unit instance. To switch between polymers and oligomers, use switch *Oligo/Poly*. Activation of a *Poly* mode sets the selected residue as an acceptor (left) terminus of the repeating unit. The reducing end (root) of the current structure becomes a donor (right) terminus automatically. Activation of an *Oligo* mode removes the hanging bonds at both termini and converts the structure to an oligomer.

Repeating Subunit Input. A subrepeat dialogue (Figure 5D) is opened when you click the subrepeat button on the top

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J. Chem. Inf. Model. 2021, 61, 4940−4948

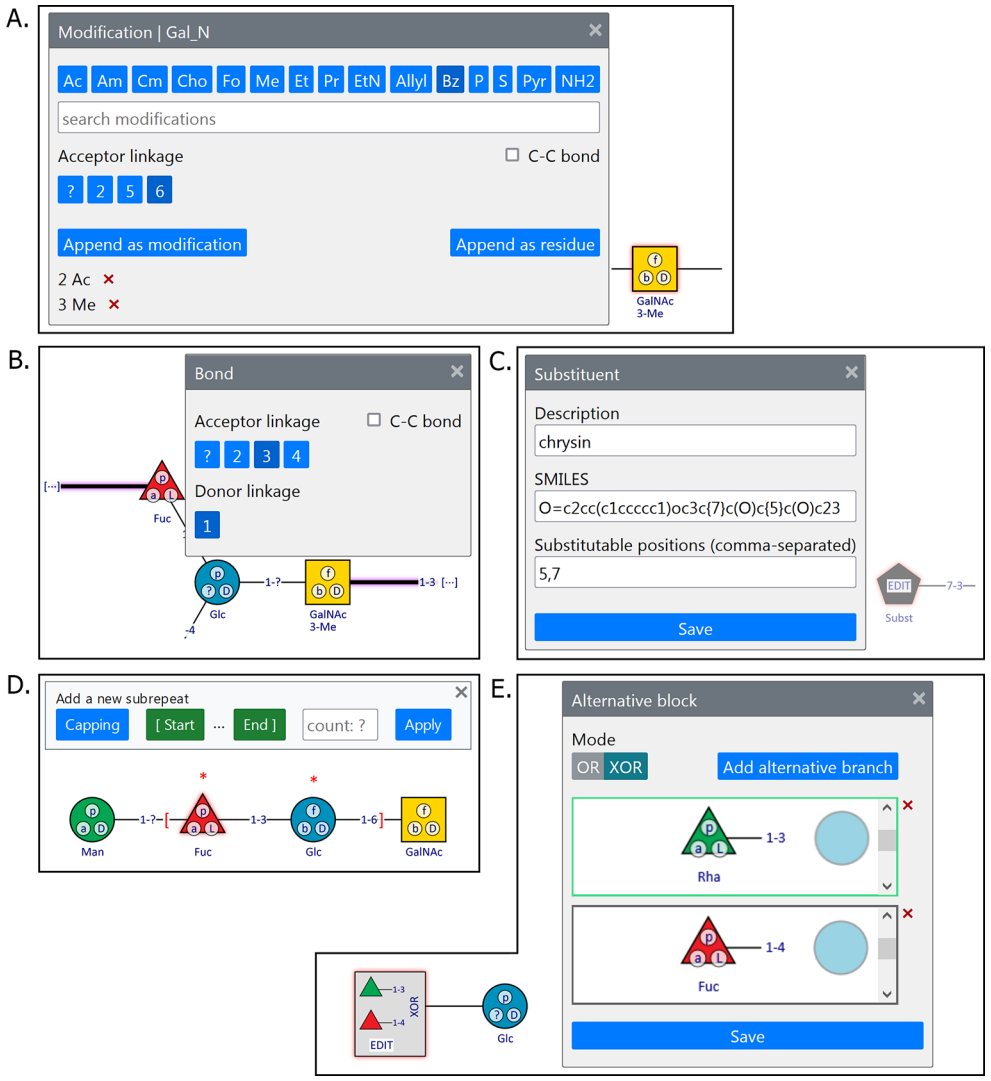


Figure 5. CSDB/SNFG Editor dialogues: (A) modifications of a residue; (B) bond properties; (C) nonvocabulary substituent; (D) repeating subunit; (E) alternative branches.

toolbar. To define the boundaries of a repeating fragment of a structure, select a residue on the sketch and click a corresponding (*Start* for the left end, *End* for the right end) button. Once a terminus is defined, residues that can be selected as another terminus are marked with a red square bracket. To change an external linkage at the left of the subrepeat (start), click Capping to loop through the allowed donors (residues or incoming bonds). To add another branch inside a subrepeat, apply changes, exit a subrepeat dialogue, select a carrier residue on the sketch, and add a branch regularly. The number of repeating units in a subrepeat should be entered in the input field (default value ? is assumed if empty). To save a subrepeat and append it to the sketch, click the *Apply* button. Once a subrepeat is added, you can only edit the number of repetitions. To change the topology of a subrepeat, delete it (click on the number of repeating units and press the *Delete* button) and create a new one. Nested subrepeats are created similarly. The Editor programmatically restricts input of subrepeats which violate the following limitations of the CSDB Linear notation:

- The whole structure, any side chain, or any repeating part cannot start with a repeating part at its right end; to encode this, extract one repeat out of a repeating part (use, e.g., /AB/n=9/AB instead of/AB/n=10/. The whole repeat should be extracted, not only its rightmost residue).
- The repeating part cannot have another nested repeating part at its left end (use, e.g., /AB/AB/n=9/C/n=20/D instead of //AB/n=10/C/n=20/D).
- There can be only one chain attached to a repeating part on the left. If the leftmost residue in the leftmost instance of a repeating part has more than one chain attached, extract it out of a repeating unit, i.e., use A[B]C/C/n=9/D instead of A[B]/C/n=10/D.

Alternative Branch Input. An alternative branch is created by clicking a corresponding button on the top toolbar. A special residue named *Fuzzy* appears on the sketch. To open the Alternative block dialogue (Figure 5E) a clickable *EDIT* control is displayed in front of the *Fuzzy* residue icon. The dialogue displays alternative branches within the target Fuzzy container and allows their addition or removal. A mini-sketch representing the alternative branch is editable in the same manner as the main sketch. The currently active branch in the block subject to commands from the toolbars and dialogues is outlined in green. Click a mini-sketch of another branch to activate it. The acceptor

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J. Chem. Inf. Model. 2021, 61, 4940−4948

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residue carrying the block of alternatives is displayed as a rightmost residue in each alternative branch as a semitransparent noneditable icon for reference only. The *OR/XOR* switch toggles the logic of a combination of alternatives (inclusive vs exclusive OR).

Quick Help and User Manual. The Editor has an integrated quick help. Quick help mode is enabled by clicking on a help button in the main toolbar. In the quick help mode, every major control and interface element gets a small question mark displayed over its corner. By clicking this question mark a popup window shows a brief description of the purpose and usage of the element.

A detailed user manual with screenshots is available at the CSDB Web site (http://csdb.glycoscience.ru/database/core/help.php?db=database&topic=usage#snfgedit). A button to open the user manual appears near the help button (Figure 2, No. 1) when the quick help mode is on.

Application. The online version of the Editor is integrated with the CSDB database interface (http://csdb.glycoscience.ru/database/core/search_struc.shtml). It is a recommended tool to build structural query terms for the CSDB database search engine. The Editor can be used to conveniently create a glycan figure illustration in raster (GIF) or vector (SVG) formats, convert it to various notations, or produce an SNFG-colored optimized 3D structure. Quick sharing of glycan structures between users is possible by copy−pasting the URL of the Editor, which contains a string representation of a currently sketched glycan as a CSDB Linear code. The Editor is available online for free download, and instructions for installation, usage, and integration into any glycoscience project are provided (http://csdb.glycoscience.ru/database/core/help.php?db=database&topic=snfgedit_ext).

METHODS

The software was written in JavaScript language (ECMAScript 2015), a modern standard for web applications. The user interface was implemented using a VueJS 2 JavaScript framework (https://vuejs.org/) and Bootstrap 4 CSS framework (https://getbootstrap.com/). Konva.js (https://konvajs.org/), a HTML5 2D-canvas JavaScript library, was applied as a graphical core for SNFG structure representation. The application is cross-platform and compatible with any web browser supporting JavaScript (ECMAScript 2015 or later versions). Integration of the Editor with CSDB online services (loading of residue and modification vocabularies, import/export operations, chemical and spatial structure visualization) is provided by a dedicated server-based web-API written in PHP 5. The CSDB/SNFG structure editor is integrated into the web interface of CSDB as one of the ways to input arbitrary glycan structures. A manual on the integration of the Editor into other projects of glycoinformatics is provided at http://csdb.glycoscience.ru/database/core/help.php?db=database&topic=snfgedit_ext.

CONCLUSION

We developed a new web-based editor for glycan and derivative structures which uses a recognized SNFG representation and supports input of polymers, repeating subunits (including nested), alternative branches, arbitrary nonvocabulary molecular fragments, chelate and carbon−carbon bonds, uncertainties in residues and linkages, and other features. The Editor features integrated visualization tools including generation of a chemical structure as a planar structural formula with stereochemical markers or a spatial molecular model. Export to string carbohydrate notations and chemical structure formats is implemented.

CSDB/SNFG Editor provides a novel export functionality of the full support of rarely used GlycoCT features (ALT, REP, NON sections). To enable a seamless support of non-carbohydrate units, an extension to GlycoCT specification was elaborated. In the extended notation, arbitrary noncarbohydrate moieties are encoded as SMILES with embedded atom numbering and referenced in RES and LIN sections similarly to units of other types. Atom numbering in SMILES allows one to encode structures having nonvocabulary constituents without loss of chemical unambiguity.

CSDB/SNFG Editor is freely available for download and can be used as a structure builder and/or visualization tool in any glycoscience project. The Editor implements a client−server model comprised of a JavaScript/HTML interface module and a group of online services available via web-API. Integration of the interface module is simple and does not require programming skills beyond basic knowledge of HTML. The used Web-API is hosted at the CSDB server, and therefore, no any specific configuration of the server environment is needed.

We hope that CSDB/SNFG Editor combines ease of usage with a wide spectrum of capabilities and will promote advances in glycoinformatics. It is purposed both for other projects (databases, web-services, and software) and for end users (glycochemists and glycobiologists).

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jcim.1c00917.

Examples of glycan structures represented by CSDB/SNFG Editor as SNFG sketches, CSDB Linear, and GlycoCT strings (PDF)

AUTHOR INFORMATION

Corresponding Author

Philip V. Toukach − *Laboratory of Carbohydrate Chemistry, Zelinsky Institute of Organic Chemistry, Russian Academy of Sciences, 119991 Moscow, Russia; Faculty of Chemistry, National Research University Higher School of Economics, 117312 Moscow, Russia;* orcid.org/0000-0001-9887-8919; Email: netbox@toukach.ru

Author

Andrei Y. Bochkov − *Laboratory of Carbohydrate Chemistry, Zelinsky Institute of Organic Chemistry, Russian Academy of Sciences, 119991 Moscow, Russia;* orcid.org/0000-0002-8798-0042

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jcim.1c00917

Notes

The authors declare no competing financial interest.

For preparation of this manuscript CSDB/SNFG Editor version 1.0.4 was used. The online version of the editor is available at http://csdb.glycoscience.ru/snfgedit/snfgedit.html. Download link and instructions for installation and usage are available at http://csdb.glycoscience.ru/database/core/help.php?db=database&topic=snfgedit_ext.

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■ ABBREVIATIONS

SNFG, Symbol Nomenclature For Glycans; CSDB, Carbohydrate Structure DataBase; SMILES, Simplified Molecular Input Line Entry Specification; WURCS, Web3 Unique Representation of Carbohydrate Structures; API, Application Programming Interface; MMFF, Merck Molecular Force Field; REStLESS, ResiduEs as SMILES and LinkagEs as SMARTS; SMARTS, SMILES Arbitrary Target Specification; InChI, International Chemical Identifier

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